Impact of Euler Angle Singularities on UAV Attitude Control: A Comparison with Quaternion-Based Formulations

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Abstract. The choice of attitude parametrization plays a fundamental role in the numerical robustness of UAV rotational control systems. Euler-angle representations remain widely used due to their intuitive roll–pitch–yaw interpretation, but they correspond to local coordinate charts of $SO(3)$ and exhibit intrinsic kinematic singularities. In contrast, quaternion-based representations provide a globally nonsingular embedding of the rotation manifold.

This work investigates how internal attitude propagation influences closed-loop UAV dynamics under a representation-independent geometric control framework. The control law is defined directly on $SO(3)$, and controller gains are selected through a fairness-based RMS equalization procedure to ensure identical nominal tracking performance across parametrizations. Under this setup, the only structural difference between the simulated cases lies in the internal propagation mechanism (Euler ZYX versus quaternion integration).

Numerical simulations drive the system toward and across the Euler singular configuration. The results show that the conditioning of the Euler kinematic mapping grows rapidly near $\theta = \pm\frac{\pi}{2}$, acting as a dynamic amplification mechanism within the feedback loop. This effect leads to increased angular velocity and control torque despite the controller itself remaining geometrically well defined. In contrast, quaternion propagation preserves smooth and bounded closed-loop behavior throughout singularity-crossing maneuvers. These results clarify how local coordinate singularities influence numerical robustness in UAV attitude control and provide a reproducible evaluation framework for comparing attitude parametrizations in reliability-critical applications.

Keywords: UAVs, attitude parametrization, geometric control, Euler singularity, quaternion, numerical conditioning

1 Introduction

Unmanned Aerial Vehicles (UAVs) are increasingly deployed in civil and industrial missions that require high levels of autonomy, reliability, and operational safety. Applications such as infrastructure inspection, mapping, and power transmission line monitoring demand accurate attitude regulation under large rotational excursions and close-proximity operation [5]. In these scenarios, the numerical robustness of the attitude representation becomes an important factor for reliable control performance.

The orientation of a rigid body evolves on the nonlinear manifold $SO(3)$, and practical implementations rely on coordinate parametrizations or embeddings of this space. Local parametrizations such as Euler angles remain widely used due to their intuitive roll–pitch–yaw interpretation and implementation simplicity. However, Euler-angle formulations suffer from intrinsic kinematic singularities, known as gimbal lock, where the mapping between body angular velocity and attitude rates becomes ill-conditioned and may degrade closed-loop behavior [4]. Global representations such as unit quaternions provide a nonsingular embedding of $SO(3)$ in S^3 and maintain smooth kinematics across the entire configuration space. For this reason, quaternion-based implementations are widely adopted in modern UAV flight controllers, including Pixhawk/PX4 platforms [6].

To address this issue, this work adopts a representation-independent geometric controller defined directly on $SO(3)$. Controller gains are selected through a fairness-based RMS equalization procedure to ensure equivalent nominal closed-loop performance. Under this framework, the only difference between the simulated cases lies in the internal attitude propagation mechanism, allowing structural effects of the parametrization to be isolated under identical closed-loop conditions. This work investigates how attitude parametrization influences closed-loop UAV rotational dynamics under a representation-independent geometric control framework. The theoretical background and modeling framework are presented in the next section.

2 Background

The orientation of a rigid body evolves on the special orthogonal group $SO(3)$, and practical implementations rely on coordinate parametrizations or embeddings of this space. Local parametrizations such as Euler angles provide an intuitive roll–pitch–yaw interpretation but are defined only on subsets of $SO(3)$. Their kinematic mapping between body angular velocity and attitude rates loses rank at specific configurations, producing the classical gimbal-lock singularity. Near these configurations, the associated Jacobian becomes ill-conditioned and numerical sensitivity increases [3].

Global representations such as unit quaternions embed $SO(3)$ into the unit sphere S^3 and avoid coordinate singularities, providing smooth kinematic evolution across the full attitude space [1]. To isolate parametrization effects, this work adopts a representation-independent geometric controller defined directly

on $SO(3)$ with gains selected through a fairness-based tuning procedure. Under this framework, performance differences arise solely from the internal propagation mechanism, which is particularly relevant for UAV maneuvers involving large attitude excursions where conditioning effects may influence closed-loop robustness [5].

3 Modeling, Geometric Control Design, and Simulation Setup

The formulation is designed to isolate the effect of attitude parametrization from the control structure, so that any performance difference can be attributed to the internal representation rather than to the controller design [7].

3.1 Rigid-Body Rotational Dynamics

The UAV rotational dynamics in the body frame are

$$I\dot{\boldsymbol{\omega}} = \boldsymbol{\tau} - \boldsymbol{\omega} \times (I\boldsymbol{\omega}), \tag{1}$$

where $I \in R^{3\times 3}$ is the inertia matrix, $\boldsymbol{\omega} \in R^3$ is the angular velocity, and $\boldsymbol{\tau} \in R^3$ is the control torque. In all simulations,

$$I = \mathrm{diag}(0.02,\ 0.02,\ 0.04)\ \mathrm{kg{\cdot}m^2}.$$

Actuator limits are modeled through component-wise saturation,

$$\tau_i \in [-\tau_{\max,i},\ \tau_{\max,i}], \qquad \tau_{\max} = (1.5,\ 1.5,\ 1.0)\ \mathrm{N{\cdot}m}.$$

3.2 Attitude Propagation

The attitude evolves on the special orthogonal group $SO(3)$ as

$$\dot{R} = R[\boldsymbol{\omega}]_\times, \tag{2}$$

where $[\cdot]_\times$ denotes the skew-symmetric operator. Two internal propagation mechanisms are considered.

Euler ZYX parametrization Using the ZYX convention,

$$R(\phi,\theta,\psi) = R_z(\psi)R_y(\theta)R_x(\phi),$$

with $\boldsymbol{\eta} = [\phi,\theta,\psi]^T$. The Euler kinematics are

$$\dot{\boldsymbol{\eta}} = E(\phi,\theta)\boldsymbol{\omega}, \tag{3}$$

where

$$E(\phi,\theta) = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \frac{\sin\phi}{\cos\theta} & \frac{\cos\phi}{\cos\theta} \end{bmatrix}.$$

This mapping becomes singular when $\cos\theta = 0$, i.e., $\theta = \pm\frac{\pi}{2}$, corresponding to the classical gimbal-lock condition [4,3].

Quaternion parametrization The quaternion state

$$\mathbf{q} = [q_0, q_1, q_2, q_3]^T, \qquad \|\mathbf{q}\| = 1$$

is propagated as

$$\dot{\mathbf{q}} = \frac{1}{2}\Omega(\boldsymbol{\omega})\mathbf{q}. \tag{4}$$

The quaternion is renormalized at each step to enforce the unit-norm constraint. Unlike Euler angles, quaternion propagation remains globally nonsingular over the full attitude space [2].

3.3 Geometric Control on $SO(3)$

To avoid representation bias, the controller is defined directly on $SO(3)$. The attitude and angular velocity errors are

$$e_R = \frac{1}{2}\mathrm{vee}(R_d^T R - R^T R_d), \tag{5}$$

$$e_\omega = \boldsymbol{\omega} - R^T R_d \boldsymbol{\omega}_d, \tag{6}$$

with $\boldsymbol{\omega}_d = \mathbf{0}$. For quantitative evaluation, the scalar attitude error

$$e_{ang} = \arccos\left(\frac{\mathrm{tr}(R_d^T R) - 1}{2}\right) \tag{7}$$

is also considered.

The control torque is

$$\boldsymbol{\tau} = -K_R e_R - K_\omega e_\omega + \boldsymbol{\omega} \times (I\boldsymbol{\omega}). \tag{8}$$

Because the controller, error definitions, and dynamics are identical in both implementations, the only difference between simulations lies in the internal attitude propagation mechanism.

Table 1. Simulation and control parameters.

Parameter	Value
Inertia matrix I	$\mathrm{diag}(0.02, 0.02, 0.04)$ kg·m^2
Torque limits $\tau_{\max}$	$(1.5, 1.5, 1.0)$ N·m
Integration method	Explicit Euler
Time step	$dt = 0.001$ s
Control gains	$K_R = 80,\ K_\omega = 2$
Test maneuvers	$30°,\ 90°,\ 120°$ pitch

3.4 Fairness-Based Gain Selection and Simulation Setup

Controller gains are selected using a fairness-based tuning procedure that equalizes nominal closed-loop performance for a 30° pitch maneuver [7]. The optimization objective is

$$J = |\mathrm{RMS}_E - \mathrm{RMS}_Q| + 0.05(\mathrm{RMS}_E + \mathrm{RMS}_Q) + 0.01(\mathrm{RMS}_{\tau,E} + \mathrm{RMS}_{\tau,Q}), \quad (9)$$

where RMS_E and RMS_Q denote the RMS scalar attitude error for Euler and quaternion implementations. The resulting gains are

$$K_R = 80, \qquad K_\omega = 2.$$

The evaluated metrics include RMS attitude error e_{ang}, control effort $\|\tau\|$, angular velocity $\|\omega\|$, and the condition number $\kappa(E)$ for the Euler case. All simulations use explicit Euler integration with time step $dt = 0.001$ s. Disturbances and sensor noise are intentionally excluded to isolate parametrization-induced conditioning effects. Three maneuvers are analyzed: 30° (nominal), 90° (near singularity), and 120° (singularity crossing).

For reproducibility, the simulation scripts and figure-generation code used in this study are publicly available at **uav-attitude-singularity-study on Github**.

4 Simulation Results and Quantitative Evaluation

This section presents the numerical results obtained under identical geometric control gains ($K_R = 80$, $K_\omega = 2$), selected through the fairness-based tuning procedure described previously. Since the controller is defined directly on $SO(3)$, both implementations share the same rotational dynamics, error definitions, and gain values. Therefore, the only structural difference between the two cases lies in the internal attitude propagation mechanism.

For the nominal 30° maneuver, Euler and quaternion propagations exhibit indistinguishable closed-loop behavior. As shown in Figure 1, the attitude response and control effort are practically identical for both representations. The Euler conditioning indicator remains close to unity, indicating that the kinematic

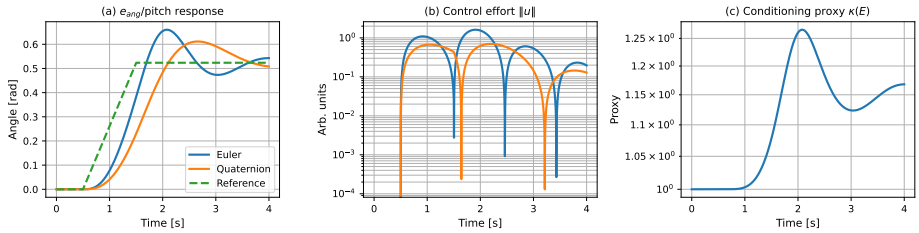

Fig. 1. Nominal 30° maneuver: attitude response, control effort, and Euler conditioning indicator.

mapping is well conditioned. This confirms that the fairness-based gain tuning successfully equalizes nominal performance and removes gain-selection bias from the comparison.

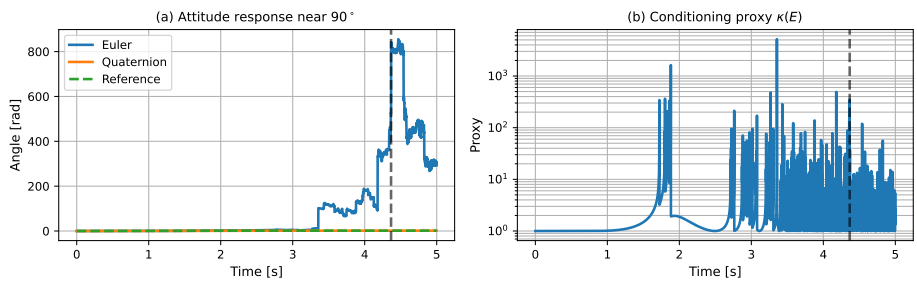

Fig. 2. Near-singularity 90° maneuver: attitude response and Euler conditioning indicator.

As the pitch approaches ±90°, the Euler kinematic matrix becomes severely ill-conditioned. Figure 2 shows a rapid growth in the conditioning indicator associated with $E(\phi, \theta)$, reaching values on the order of 10^{23} (Table 2). Although the control law remains unchanged and globally well defined on $SO(3)$, the mapping $\dot{\boldsymbol{\eta}} = E(\phi, \theta)\boldsymbol{\omega}$ amplifies small body angular velocities into large coordinate-rate variations. This amplification propagates through the feedback loop, increasing angular velocity and control effort. In contrast, quaternion propagation remains well conditioned throughout the maneuver.

The effect becomes more pronounced when the trajectory crosses the singular configuration. In the 120° maneuver, the Euler parametrization exhibits clear dynamic amplification: the conditioning indicator increases rapidly and pronounced peaks appear in both angular velocity magnitude and control effort. As illustrated in Figure 3, this degradation occurs despite the controller

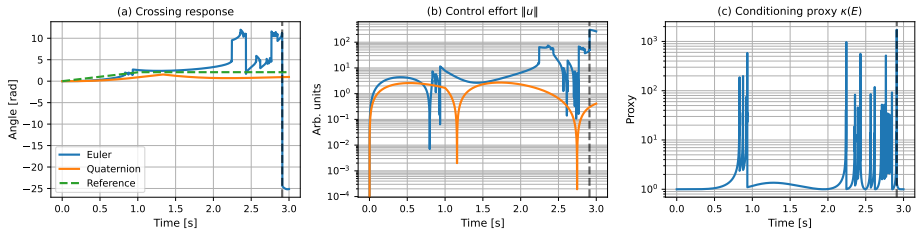

Fig. 3. Singularity-crossing 120° maneuver: attitude response, control effort, and Euler conditioning indicator.

remaining unchanged, indicating that the effect originates solely from the internal propagation mapping. In contrast, quaternion propagation preserves smooth and bounded closed-loop behavior throughout the maneuver.

Table 2. Key closed-loop metrics under identical geometric control gains.

Case	Model	RMS e_{ang}	Max τ	Max $\kappa(E)$
Nominal 30°	Euler	0.006505	0.285821	1.73
Nominal 30°	Quaternion	0.006505	0.285821	–
Near 90°	Euler	0.014283	0.571621	2.67×10^{23}
Near 90°	Quaternion	0.014283	0.571621	–
Crossing 120°	Euler	0.030058	1.143074	8.35×10^{4}
Crossing 120°	Quaternion	0.030058	1.143074	–

Table 2 summarizes the quantitative results. Three observations emerge. First, nominal performance is identical for both parametrizations, confirming that the fairness condition is satisfied. The results confirm that the observed degradation in the Euler case originates from the analytical structure of the kinematic mapping $E(\phi, \theta)$. As $\theta \to \frac{\pi}{2}$, the terms proportional to $\frac{1}{cos(\theta)}$ dominate the mapping and the number of conditions grows rapidly. This introduces a multiplicative amplification between body angular velocity and coordinate rates, which propagates through the feedback loop and increases the control effort. Quaternion propagation remains globally nonsingular and avoids this coordinate-induced amplification, preserving smooth closed-loop behavior.

5 Conclusion

This work investigated the influence of attitude parametrization on closed-loop UAV rotational dynamics under a representation-independent geometric control framework. The controller was defined directly on $SO(3)$ and identical gains were selected using a fairness-based RMS equalization procedure, ensuring equivalent

nominal closed-loop performance across parametrizations. Under this setup, the only structural difference between the simulated cases lies in the internal attitude propagation mechanism.

The results show that Euler-angle singularities can act as dynamic amplification mechanisms when embedded in closed-loop systems. As the pitch angle approaches $\pm 90°$, the conditioning of the Euler kinematic mapping increases rapidly, introducing numerical ill-conditioning in the internal state update. Although the geometric controller remains well posed, this ill-conditioning propagates through the feedback loop, resulting in amplified angular velocity and increased control effort. In contrast, quaternion propagation maintains globally well-conditioned kinematics and preserves smooth attitude evolution and bounded control effort even during singularity-crossing maneuvers.

These findings demonstrate that, even when control is formulated intrinsically on $SO(3)$, the internal attitude representation can significantly influence numerical robustness in closed-loop dynamics. This highlights that coordinate singularities may act as hidden amplification mechanisms within feedback systems.

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